

Clinical Risk Assessment Report: Patient 64

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 10.3% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.47x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 3.4%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

No major risk-elevating features observed in the top drivers.

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) Value: -0.55 [Avg (27th %ile)] (-3.4%)

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) Value: -0.29 [Avg (30th %ile)] (-1.1%)

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) Value: -0.43 [Bottom <1%] (-0.6%)

Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

Short Run High Gray-Level Emphasis (CT) (GLRLM_SRHGE.CT) Value: 0.02 [Avg (42th %ile)] (-0.6%)

Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.

Squamous Cell Carcinoma Antigen (SCC_Ag) Value: 0.9 [Avg (74th %ile)] (-0.5%)

Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) Value: -0.71 [Bottom 11%] (-0.5%)

Points to continuous, elongated bands of high-density or highly active tumor tissue.

PET Radiomic Metabolic Feature (GLZLM_ZLNU.PET) Value: -0.49 [Bottom <1%] (-0.4%)

Metabolic PET feature reflecting glucose avidity and cellular metabolic rate.

Small Zone High Gray-Level Emphasis (PET) (GLZLM_SZHGE.PET) Value: -0.54 [Bottom 10%] (-0.3%)

Indicates multiple small areas of intense density or metabolism, a marker of erratic and aggressive cellular proliferation.

Machine Learning Feature Attribution (SHAP Decision Path)

